



Project Information

Project Name:	{BMI calculator and Fitness tracker}
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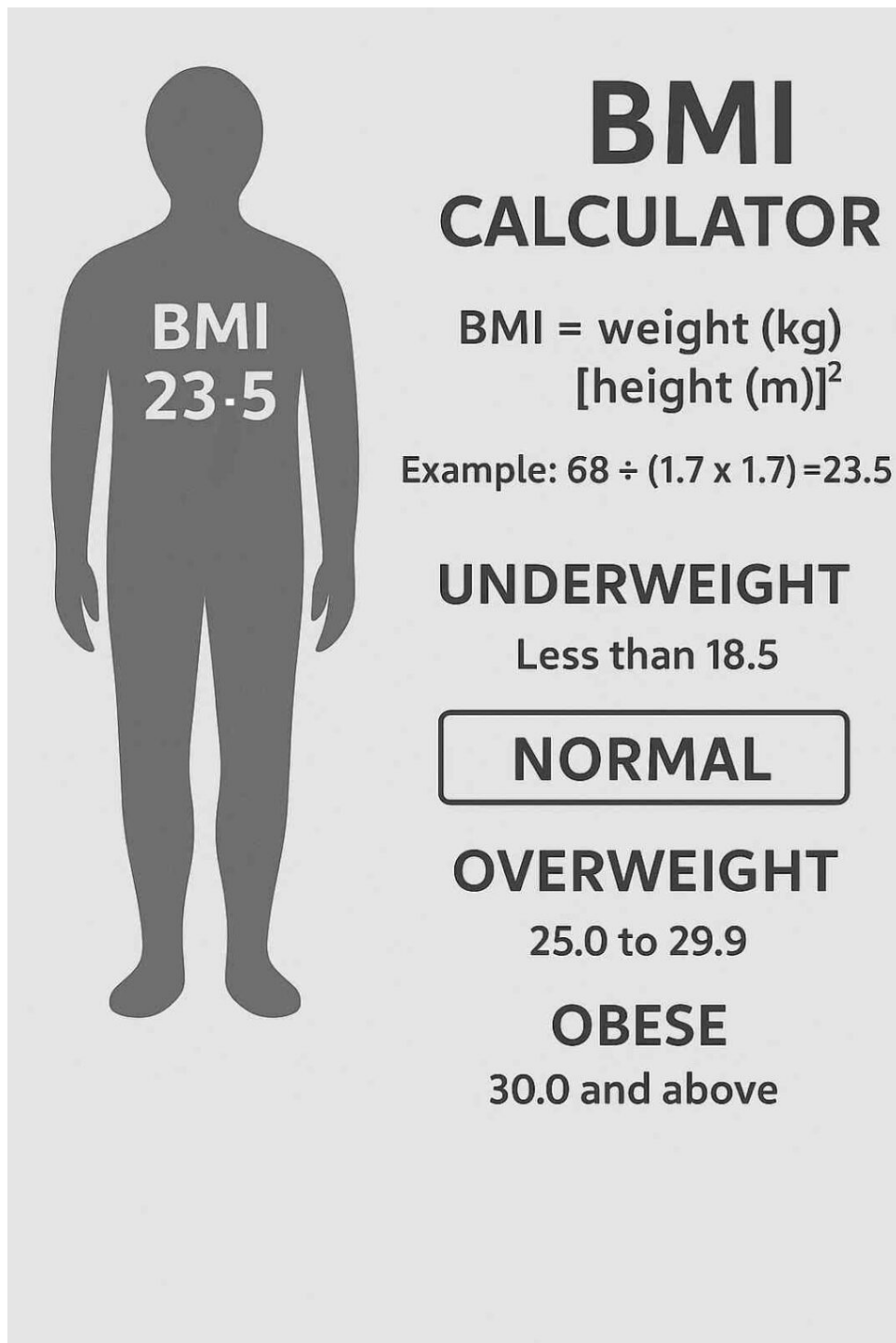
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1. Introduction

This project is a BMI Calculator developed using HTML, CSS, and JavaScript. The objective was to create a functional and user-friendly web application that allows users to determine their Body Mass Index by inputting age, gender, height, and weight.

The interface is designed with a clean, responsive layout, providing clear feedback on the BMI result and offering relevant health suggestions based on the calculation.



2. HTML Overview

The HTML structure provides the skeleton for the BMI Calculator. It uses semantic containers to organize the input form, the calculation triggers, and the results display sections. The form fields are structured to be intuitive for the user.

-This image shows the basic structure of the web page, including the document setup, linked CSS file, and the main container. It also contains the form inputs for age, gender, and height, which allow the user to enter data before calculating BMI.

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <!-- basic page setup -->
5    <meta charset="UTF-8">
6    <meta name="viewport" content="width=device-width, initial-scale=1.0">
7    <title>BMI Calculator</title>
8    <link rel="stylesheet" href="style.css">
9  </head>
10 <body>
11   <!-- this is the main box that holds everything -->
12   <div class="box">
13     <!-- main title -->
14     <h1>BMI CALCULATOR</h1>
15     <!-- each row has a label on the left and input on the right -->
16     <div class="form-row">
17       <label for="age">Age</label>
18       <input type="number" id="age" placeholder="21">
19     </div>
20     <!-- gender row -->
21     <div class="form-row">
22       <label for="gender">Gender</label>
23       <select id="gender">
24         <option value="">Choose gender</option>
25         <option value="Male">Male</option>
26         <option value="Female">Female</option>
27       </select>
28     </div>
29     <!-- height row -->
30     <div class="form-row">
31       <label for="height">Height</label>
32       <input type="number" id="height" placeholder="170">
33     </div>
34     <!-- weight row -->
35     <div class="form-row">
36       <label for="weight">Weight</label>
37       <input type="number" id="weight" placeholder="75">
38     </div>
```

-This image shows the continuation of the page structure, including the calculate button and the section where the BMI result and messages are displayed. It also includes the colored BMI category boxes that help users understand their result visually.

```
39     <!-- button to start the calculation -->
40     <button id="calculateBtn">Calculate</button>
41     <!-- this section shows the final result -->
42     <div class="result">
43         <p id="infoResult">Info: -</p>
44         <p id="bmiResult">Your BMI: -</p>
45         <p id="bmiMessage">Category: -</p>
46         <p id="suggestionResult">Suggestion: -</p>
47     </div>
48     <!-- these are the color boxes at the bottom like the picture -->
49     <div class="bmi-cards">
50         <div class="card blue">
51             <span>UNDERWEIGHT</span>
52             <strong>&lt; 18.5</strong>
53         </div>
54         <div class="card green">
55             <span>HEALTHY</span>
56             <strong>18.5 - 24.9</strong>
57         </div>
58         <div class="card orange">
59             <span>OVERWEIGHT</span>
60             <strong>25 - 29.9</strong>
61         </div>
62         <div class="card red">
63             <span>OBESE</span>
64             <strong>&ge; 30</strong>
65         </div>
66     </div>
67 </div>
68 CodeSnap
69 <!-- javascript file -->
70 <script src="script.js"></script>
71 </body>
72 </html>
```


3.CSS Overview

The styling is crucial for the modern look and feel of the application. We utilized CSS Grid for the layout, defined specific color palettes for the BMI ranges (Underweight, Healthy, Overweight, Obese), and added responsive breakpoints to ensure compatibility across different screen sizes.

-These images shows the main styling of the webpage, including global settings like margins, fonts, and colors, along with the layout of the main container and form elements. They also define the design of input fields, buttons, and sections to create a clean and organized user interface.

```
1  /* page background and basic font */
2  body {
3      font-family: Arial, sans-serif;
4      background-color: #f9f4e9;
5      margin: 0;
6      padding: 10px;
7      color: #333a45;
8  }
9  /* the main area */
10 .box {
11     max-width: 660px;
12     margin: 8px auto;
13     padding: 10px 14px 8px;
14 }
15 /* main title */
16 h1 {
17     margin-top: 20;
18     margin-bottom: 30px;
19     text-align: center;
20     font-size: 23px;
21     letter-spacing: 1px;
22 }
23 /* normal text style */
24 p {
25     font-size: 14px;
26     margin: 7px 0;
27 }
28 /* each form line */
29 .form-row {
30     display: grid;
31     grid-template-columns: 170px 1fr;
32     gap: 14px;
33     align-items: center;
34     margin-bottom: 9px;
35 }
36 /* labels on the left side */
37 label {
38     display: block;
39     font-size: 19px;
40     font-weight: bold;
41 }
42 /* text boxes and dropdowns */
43 input,
44 select {
45     width: 100%;
46     padding: 8px 12px;
47     box-sizing: border-box;
48     border: 2px solid #d3d3d3;
49     border-radius: 10px;
50     background-color: white;
51     font-size: 18px;
52     color: #333a45;
53 }
54 /* calculate button */
55 button {
56     display: block;
57     margin: 12px auto 0;
58     width: 165px;
59     padding: 9px;
60     border: none;
61     border-radius: 10px;
62     background-color: #333a45;
63     color: white;
64     font-size: 16px;
65     cursor: pointer;
66 }
67 /* result area under the button */
68 .result {
69     margin-top: 14px;
70     padding: 10px 12px;
71     background-color: rgba(255, 255, 255, 0.5);
72     border-radius: 10px;
73 }
74 /* boxes at the bottom */
75 .bmi-cards {
76     display: grid;
77     grid-template-columns: repeat(4, 1fr);
78     gap: 8px;
79     margin-top: 12px;
80 }
```

-These images show the detailed styling of individual components such as the BMI result section and category boxes. They also define colors, spacing, and visual elements that improve readability and make the interface more user-friendly.

```
81  /* each color box */
82  .card {
83      min-height: 82px;
84      border-radius: 10px;
85      color: white;
86      display: flex;
87      flex-direction: column;
88      justify-content: center;
89      align-items: center;
90      text-align: center;
91      padding: 10px 6px;
92  }
93  /* small title inside the card
94  .card span {
95      font-size: 12px;
96      font-weight: bold;
97      margin-bottom: 9px;
98  }
99  /* bmi range text */
100 .card strong {
101     font-size: 16px;
102 }
103 /* underweight color */
104 .blue {
105     background-color: #45a0dc;
106 }
107 /* healthy color */
108 .green {
109     background-color: #4bbc63;
110 }
111 /* overweight color */
112 .orange {
113     background-color: #f7a329;
114 }
115 /* obese color */
116 .red {
117     background-color: #e94a43;
118 }
```

```
119 /* simple phone layout */
120 @media (max-width: 700px) {
121     /* stacking the form for smaller screens */
122     .form-row {
123         grid-template-columns: 1fr;
124         gap: 8px;
125     }
126     /* smaller labels on phone */
127     label {
128         font-size: 16px;
129     }
130     /* smaller text inside inputs on phone */
131     input,
132     select {
133         font-size: 16px;
134     }
135     /* cards become 2 per row on phone */
136     .bmi-cards {
137         grid-template-columns: repeat(2, 1fr);
138     }
139     /* smaller title on phone */
140     h1 {
141         font-size: 20px;
142         margin-bottom: 14px;
143     }
144 }
145
```

4. JavaScript Overview

The JavaScript logic performs the critical calculations. Upon clicking the "Calculate" button, the script gathers user input values, converts them into valid metrics, calculates the BMI, and dynamically updates the page content with the health category and personalized advice.

-This image shows the main JavaScript logic that handles user input and calculates the BMI value based on height and weight. It also includes basic validation to ensure the entered data is correct before performing the calculation.

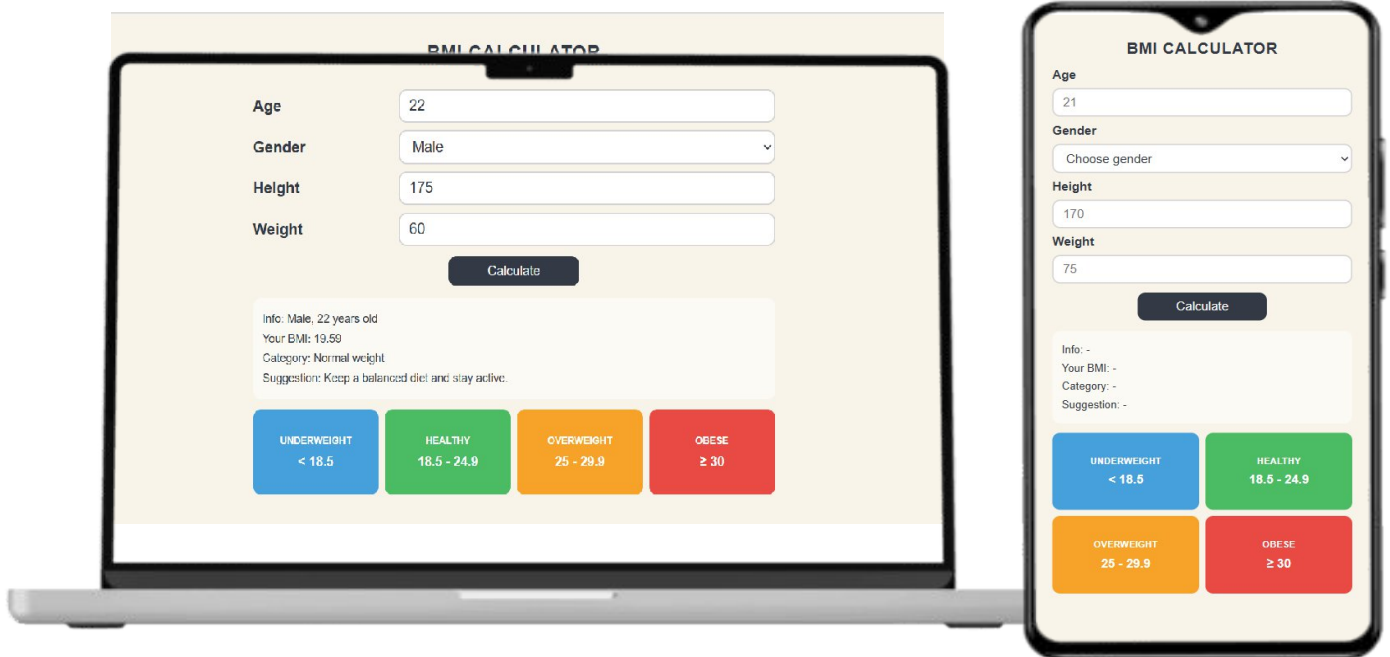
```
1  // getting all the elements from the page
2  var ageInput = document.getElementById("age");
3  var genderInput = document.getElementById("gender");
4  var heightInput = document.getElementById("height");
5  var weightInput = document.getElementById("weight");
6  var button = document.getElementById("calculateBtn");
7  var infoResult = document.getElementById("infoResult");
8  var bmiResult = document.getElementById("bmiResult");
9  var bmiMessage = document.getElementById("bmiMessage");
10 var suggestionResult = document.getElementById("suggestionResult");
11 // when the button is clicked this function will run
12 button.addEventListener("click", function () {
13     // getting the values the user typed
14     var age = parseFloat(ageInput.value);
15     var gender = genderInput.value;
16     var heightCm = parseFloat(heightInput.value);
17     var weightKg = parseFloat(weightInput.value);
18     // checking if the user entered all the needed info
19     if (!age || age <= 0 || !gender || !heightCm || !weightKg || heightCm <= 0 || weightKg <= 0) {
20         infoResult.textContent = "Info: -";
21         bmiResult.textContent = "Your BMI: -";
22         bmiMessage.textContent = "Category: Please fill in all fields correctly.";
23         suggestionResult.textContent = "Suggestion: -";
24         return;
25     }
```


-This image shows the part of the JavaScript code responsible for displaying the BMI result and updating the interface dynamically. It assigns the calculated category and message, then shows them to the user in a clear and interactive way.

```
26 // converting cm to meters because the bmi formula uses meters
27 var heightM = heightCm / 100;
28 // squaring the height
29 var heightSquared = heightM * heightM;
30 // bmi formula: weight divided by height squared
31 var bmi = weightKg / heightSquared;
32 var roundedBmi = bmi.toFixed(2);
33 var category = "";
34 var suggestion = "";
35 // this decides which bmi range the user is in
36 if (bmi < 18.5) {
37     category = "Underweight";
38     suggestion = "Try eating more calories and doing weight lifting.";
39 } else if (bmi < 25) {
40     category = "Normal weight";
41     suggestion = "Keep a balanced diet and stay active.";
42 } else if (bmi < 30) {
43     category = "Overweight";
44     suggestion = "Try to eat healthier and do more regular exercise.";
45 } else {
46     category = "Obese";
47     suggestion = "Try to reduce calories, move more, and ask a doctor for advice if needed.";
48 }
49 // showing all the results on the page
50 infoResult.textContent = "Info: " + gender + ", " + age + " years old";
51 bmiResult.textContent = "Your BMI: " + roundedBmi;
52 bmiMessage.textContent = "Category: " + category;
53 suggestionResult.textContent = "Suggestion: " + suggestion;
54 });
55
```

5.Final Webpage Overview

Below is the final functional interface of the BMI Calculator project.



In the end I would love to thank:

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Dr. Alaa Youssef

For all the Precious informations, that helped me making this project.